



# SUDHARSHAN RAM JAYARAMAN

DATA ENGINEER INTERN

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## PROFILE

Master's student in Computational Science and Engineering with strong focus on Machine learning and data analytics. Having more than 2 years of experience in Big data analytics and design optimization in industrial environments. Seeking an master's thesis opportunity in the field of AI, data engineering and design optimization.

## EDUCATION

- **Master of Science in Computational Science and Engineering** **Munich, Germany**  
*Technical University of Munich* **GPA: 2,3**  
**Coursework:** Deep Learning, Machine Learning, Parallel Programming, Visual Data Analytics. *2024 – Present*
- **Bachelor of Technology in Mechanical Engineering** **Coimbatore, India**  
*Amrita Vishwa Vidyapeetham* **GPA: 1,4**  
**Coursework:** Finite Element Methods (FEM), CFD, Thermodynamics *2018 – 2022*

## WORK EXPERIENCE

**Hilti Entwicklungsgesellschaft GmbH** **Germany**  
*Intern - Data Engineer* *April 2026 – Present*

- Developed a state error propagation model for BLE-based IoT tracking systems, and the model projection **10 %** error magnification in gateway-to-cloud data inconsistencies.
- Automated asset-location diagnostics using **PySpark**, **Microsoft Fabric**, and API-based data ingestion for customer issue investigation.
- Designed a **session-weight based probabilistic algorithm** to predict asset locations from IoT data.
- Built end-to-end **ETL pipelines**, data warehousing workflows, and Power BI dashboards.

**Hilti Entwicklungsgesellschaft GmbH** **Germany**  
*Working Student - Data Analyst* *May 2025 – Februray 2026*

- Performed data wrangling on **Big data** battery logs using Pandas, and Pyspark to ensure high data quality.
- Implemented **ETL architecture** in Fabric for data flow and SQL queried to split the data into sub lakehouses.
- Implemented **DBSCAN clustering** to identify voltage anomalies using Scikit-learn.
- **Product owner of Sercomm tool** to Streamline data ingestion with excel macros, and automated workflows.
- Experience with **JIRA Kanban** model and Microsoft office 365 for project management.
- The tool and pipeline improved process efficiency by **24%** with automation and process optimization.

**Tata Technologies** **India**  
*Design Engineer* *Jun 2022 – Oct 2023*

- Design of chassis components (brackets and calipers) for Volvo XC series using Siemens NX and CATIA V5.
- Product ownership in **Pad Wear Indicator (PWI)** with JIRA Agile methods , contributing to **7%** profit in project.
- Design packaging of cable routing, clip assembly and design instance management in Teamcenter.
- **Built data automation pipeline** for scan data analysis.

## KEY PROJECTS

### End-to-End Banking Data Engineering Platform (Snowflake, dbt, Airflow, Kafka, Python, Docker, GitHub CI/CD)

- Designed banking data pipeline using **Kafka, Debezium, and PostgreSQL** to capture and stream transactional data through CDC.
- Orchestrated automated data ingestion and processing workflows with **Apache Airflow**, loading data into a **Snowflake** cloud warehouse (medallion architecture).
- Developed **dbt** transformation frameworks, including staging, fact, and dimension models with **SCD Type-2** history tracking.
- Developed **Power BI** dashboards for **KPI reporting** and automated **CI/CD**, testing, and data quality workflows.

### Machine Learning in crowd dynamics for cooperative Robot (Python, ML, git)

- Generated a time series Big data dataset from decentralized robot simulations, to analyze **crowd movement patterns** and evaluate AI frameworks (ResNet) for dynamic trajectory prediction..
- Performed **PCA analysis** with Scikit-learn retaining **95%** of the variance in high dimensional space.
- **Modelled Ridge Regression and RBF models** (using K-Means) to predict robots system states.
- Implemented and optimized both MLP and ResNet neural network architectures using **BOHB optimization**.

### Parallel CFD Solver Development (C++, MPI, OpenMP, Git)

- Developed a 2D incompressible Navier–Stokes CFD solver in **C++** using finite-difference schemes on a staggered grid.
- Implemented **MPI domain decomposition**, OpenMP, and GPU acceleration for scalable simulations.
- Extended the solver with **Multigrid** methods (V-cycle and W-cycle) to accelerate pressure convergence and improve **computational efficiency**.
- Visualized and validated simulation results (velocity, pressure fields) using ParaView.

### Visual data analysis of sales trend in US market (Tableau, Data Analytics)

- Developed interactive **Tableau dashboards** to analyze U.S. sales trends.
- Performed clustering and trend analysis to identify patterns and evaluate sales performance across markets.
- Analyzed the **impact of visual channels** on human perception, applying principles of visual data analytics.
- Assessed impact of visual channel selection on **human perception** and **attention** with human factor engineering.

## TECHNICAL SKILLS

|                          |                                                              |
|--------------------------|--------------------------------------------------------------|
| <b>Languages:</b>        | Python, C++, SQL, MATLAB                                     |
| <b>Data &amp; AI:</b>    | NumPy, Pandas, Scikit-learn, PySpark.                        |
| <b>Data Engineering:</b> | Microsoft Fabric, Databricks, MySQL, Airflow, Kafka, AWS S3. |
| <b>Other Tools:</b>      | Github, Docker, JIRA, Office 365, Excel, Tableau, Power BI   |

## CERTIFICATIONS

- **Data Engineering:** pySpark, Snowflake (Coursera); Databricks Fundamentals.
- **Machine Learning:** Applied Machine Learning in Python (Coursera); 100 Days of Code (Python).
- **Generative AI:** Prompt Engineering; Data Analysis with LLM.
- **Database:** SQL Intermediate Certification (HackerRank).

## LANGUAGES

**English:** C1 (Professional Working Proficiency)  
**Tamil:** Native

**German:** B2 (Independent User)